

8.2.55

AI25BTECH11017-BALU

Question:

Equation of the hyperbola with eccentricity $\frac{3}{2}$ and foci at $(\pm 2, 0)$ is _____.

Solution:

Let us solve the given equation theoretically and then verify the solution computationally
According to the question,

$$e = \frac{3}{2} \quad \mathbf{F}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \mathbf{F}_2 = \begin{pmatrix} -2 \\ 0 \end{pmatrix} \quad (0.1)$$

$$\mathbf{n} = \mathbf{F}_1 - \mathbf{F}_2 = \begin{pmatrix} 4 \\ 0 \end{pmatrix} \quad (0.2)$$

$$\mathbf{n}^T \mathbf{x} = c \quad (0.3)$$

hyperbola equation is

$$\mathbf{x}^T \mathbf{v} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \quad (0.4)$$

$$\mathbf{v} = \|\mathbf{n}\|^2 \mathbf{I} - e^2 \mathbf{n} \mathbf{n}^T \quad (0.5)$$

$$\mathbf{u} = ce^2 \mathbf{e}_1 - \mathbf{F} \quad (0.6)$$

$$f = \|\mathbf{n}\|^2 \|\mathbf{F}\|^2 - c^2 e^2 \quad (0.7)$$

$$\mathbf{u} = \frac{\mathbf{F}_1 + \mathbf{F}_2}{2} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (0.8)$$

$$\mathbf{v} = \begin{pmatrix} -20 & 0 \\ 0 & 16 \end{pmatrix} \quad (0.9)$$

$$f = 60 \quad (0.10)$$

hence the required hyperbola equation is

$$\mathbf{x}^T \begin{pmatrix} -20 & 0 \\ 0 & 16 \end{pmatrix} \mathbf{x} + 60 = 0 \quad (0.11)$$

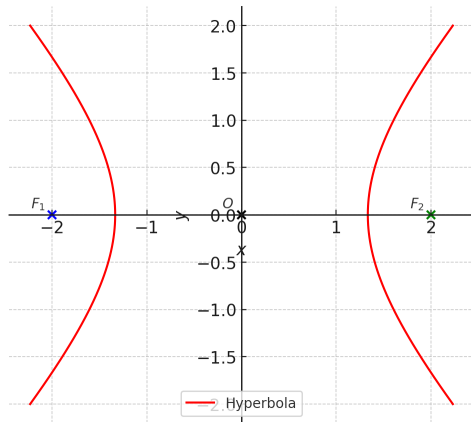


Fig. 0.1